

Life Is On

Schneider
Electric



DNF7

Metal-Clad Switchgear

Air Insulated Switchgear





As a global specialist in energy management with operations in more than 100 countries, Schneider Electric offers integrated solutions across multiple market segments, including leadership positions in energy and infrastructure, industrial processes, building automation, and data centres/networks, as well as a broad presence in residential applications. Focused on making energy safe, reliable, and efficient, the company's 160,000 plus employees achieved sales of more than 24.7 billion Euros in 2016, through an active commitment to help individuals and organisations make the most of their energy.

Schneider Electric in China

Since the establishment of its first joint-venture plant in Tianjin in 1987, Schneider Electric has established a strong foothold in the market for over 30 years, and has grown together with the Chinese economy. The most outstanding contribution made by Schneider Electric in that period was to bring circuit breaker technology into China, replacing the traditional fuse, and eventually setting new standards for breakers in China. In early 90s, the Clipsal by Schneider Electric, launched its switch panel in China, ending the Chinese history of long-term using lamp cord.

Schneider Electric's huge amount of investment constituted a strong support for China's economic construction. In the meantime, Schneider Electric provided advanced product support and sophisticated technical services for Chinese economic development: the company's industrial products, such as the low-voltage apparatuses, drives, and contactors, were extensively used in China's domestic economic development, therefore promoting the country's industrialization.

Till now, Schneider Electric had established 77 offices, 26 factories, six distribution centers, one learning institute, three research and development centers, one laboratory, 500 distributors and a nation-wide sales network. Schneider Electric currently employs 26,000 staff in China, and helps create thousands of other jobs through its partners and distributors.

Schneider Electric EcoStruxure™ Solutions

With the professional know-how in multi markets we operate and the close care of our customers, as well as our best practices in energy management, Schneider Electric has grown up from a provider of best-in-class products into an integrated solution provider. This year, we launched EcoStruxure™, an architectural approach which unites Schneider Electric's unique expertise in power, datacentres, process and machines, building control, and physical security to enable intelligent energy management solutions for customers seeking to optimise energy efficiencies across multiple domains of their business. By providing our customers with clear and comprehensive reference architectures across key environments and applications, we intend to reduce inefficiencies and save energy up to 30%.



General Description

Whether you are planning new switchgear installations or upgrading existing ones, you need high safety and reliable switchgear. The latest range of air insulated metal-clad switchgear DNF7, which is designed for indoor application, based on Schneider Electric T&D's Eco-Design concept, is a cost-effective and reliable switchgear used in 36kV and 40.5kV systems, and can totally satisfy your requirements.

Compactibility and Robustness

DNF7 metal-clad switchgear is constructed by aluminum-zinc sheet metal in a double bending process and riveting fixing technology, to form an extremely rigid construction

DNF7 is assembled by means of the riveting nuts, high strength bolts and steel core rivets connection, and no holes are needed between panels to make installation easy, improve the production efficiency.

DNF7 metal clad switchgear consists of four separated compartments, including Busbar compartment, Circuit Breaker compartment, Cable compartment and Low voltage compartment, and it is a LSC2B service continuity, PM partition class switchgear.



Interchangeability and Versatility

DNF7 can be equipped with Schneider Electric's withdrawable HVX (Vacuum Circuit Breaker) and FP (Gas Insulated Circuit Breaker).

Circuit breakers with the same rating are interchangeable irrespective of the switching technology.

DNF7 provides Middle Rolling and Floor Rolling two different withdrawable types, can meet all our customers' technical requirements and operating features.

Versatility design make DNF7 provide a complete distribution and control solution for our customer, typical scheme including feeder, incomer, metering, measuring, coupler, etc.

Customer benefits

- Space saving
- Safe and user-friendly
- Installation Convenient
- Fully internal arc tested
- Reduced installation and maintenance costs



Safety and Reliability

Pressure release devices on the top of panel can be opened automatically to the rear side of DNF7 in case of overpressure which caused by internal fault, the overpressure will be released upward, protect the operating person and prevent the extending of the fault to adjacent compartments and switchgears.

Effective and reliable interlocking system to prevent the mal-operation.

All operations are carried out from the front of the DNF7, with the door closed, effective and reliable interlocking system prevents mal-operation, providing a safer environment for the operators, Operating Rod design can let customer switch off the circuit breaker in emergency situation, it also includes independently operated padlockable shutters, visual verification of the earthing switch position and optional capacitive voltage indication, all these ensuring maximum operator safety.

Fully internal arc tested in CESI and XIHARI according to IEC 62271-200 standards.

DNF7 is designed to meet the latest specifications of IEC, Chinese "GB" and "DL" as well as many other national specifications.

Application Field

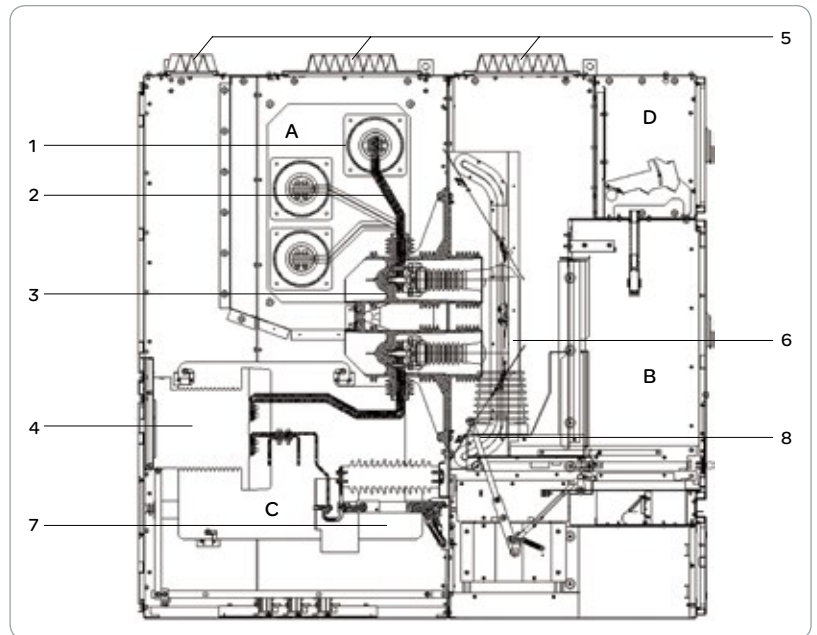
DNF7 can be used in Power Plant, Wind Farm, Iron and Steel works, Petroleum and Chemical industry, Wind Farm, Textile, Paper and Food Industries, etc.



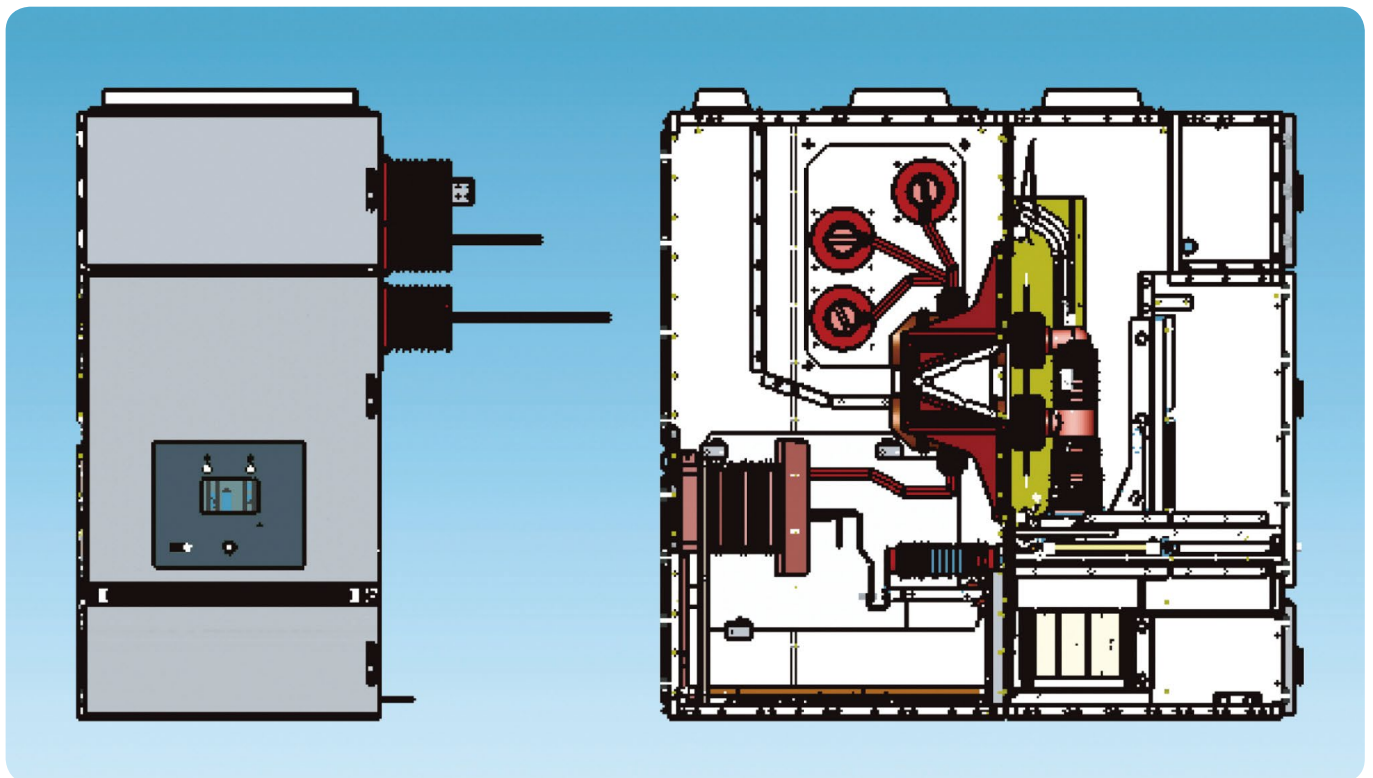
DNF7 Metal-Clad Switchgear

DNF7 Section View

- A. Busbar Compartment
- B. Circuit Breaker Compartment
- C. Cable Compartment
- D. Low Voltage Compartment
- 1. Bushing
- 2. Busbar
- 3. Spout
- 4. CT
- 5. Pressure release devices
- 6. VCB(HVX)
- 7. ESW earthing switch
- 8. Shutter system



DNF7 (Middle Rolling)



DNF7 Metal-Clad Switchgear



DNF7 includes the Panel itself and the Withdrawable Part (VCB HVX)

Busbar Compartment, Circuit Breaker Compartment, Cable Compartment, Low Voltage Compartment and these four function units form the panel itself.

Busbar compartment

Both main busbar and branch busbar are covered with heat shrink sleeves to ensure the reliable insulation. Bushings can prevent the fault spread to adjacent panels, stainless steel plate which bushings are installed on can limit the eddy current effectively, to prevent the temperature rise.



Circuit Breaker Compartment

Both Vacuum and SF6 circuit breaker can be equipped with DNF7.

Automatic safety metal shutters between circuit-breaker and bus-bar/cable compartments make the isolating bushings are completely covered and form the efficient isolation, meantime, due to the linkage between the two shutters, the shutter can be locked when it's live to protect the operator against the live parts during test and maintenance.

Inspection window on the door make the position of withdrawable circuit breaker and the charging state of the circuit breaker visible.

All operating and moving of the circuit breaker behind a closed door, this ensures the operator can get the maximal protection in case of the internal fault.



Cable compartment

Large space and enough height for multiple cable connections, surge arresters, current transformers, high speed earthing switch with making capacity are installed in the cable compartment.

Low voltage compartment

Secondary devices like MCB, relay, terminals, etc, will be installed in Low voltage compartment. External cables supplied by the customer will be routed upwards to this compartment through the left-hand cable duct (metal enclosed) in the inside of the panel as well.



Technical Data

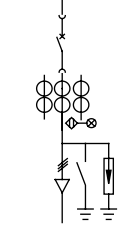
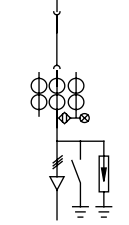
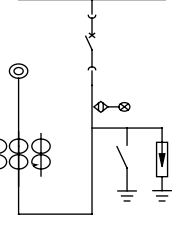
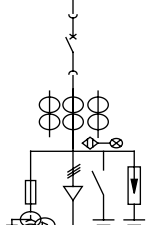
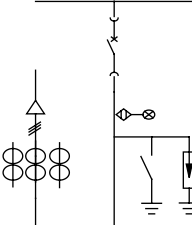
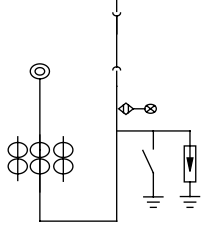
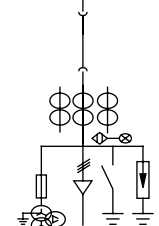
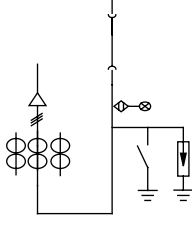
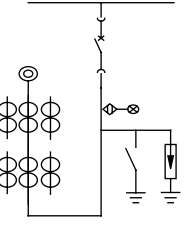
DNF7			
		IEC	CNS
Rated voltage	kV	36	40.5
Rated lightning impulse with stand voltage	kV	170	185
Rated power frequency with stand voltage	kV	70	95
Rated frequency	Hz	50/60	50
Rated normal current	A	1250/1600/2000/2500/3150 ⁽¹⁾	1250/1600/2000/2500/3150 ⁽¹⁾
Rated short-time current	Hz	50/60	50
Rated peak withstand current (peak value)	kA	65/82	63/80
Rated short-circuit making current	kA	65/82	63/80
Auxiliary Voltage	V	DC ⁽³⁾ 24,48,60,110,125,220	DC 110,220
		AC 110/120,220/230	AC 110,220
Internal arc classification	AFLR 31.5kA (1s)		
Protection degree of the enclosure	IP4X (2)		
Protection degree of the circuit-breaker compartment door open	IP4X		
Standard dimensions (W×D×H)	mm		1200(1400)×2505×2400
Max. Weight with HVX without packaging	kg	1800	

- 1) Forced cooling ventilation is required
- 2) IP41 for option
- 3) Special DC voltage on required

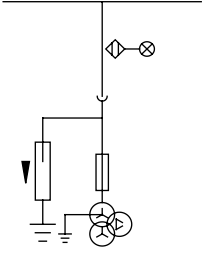
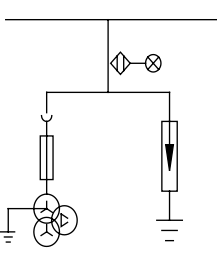
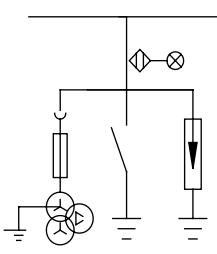
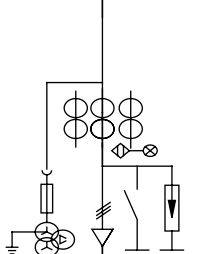


DNF7 Metal-Clad Switchgear

Primary Scheme

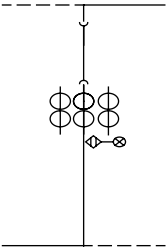
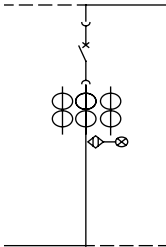
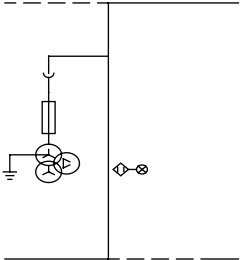
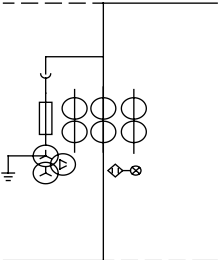
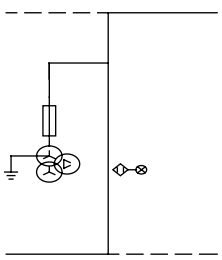
No.		I/F01	I/F02	I/F03	I/F04	I/F05
Incomer / Feeder						
Rated Current	HVX (VCB)	630-2500A		630-3150A	630-2500A	630-3150A
	FP (SF6)	630-2500A		630-2000A	630-2000A	630-2000A
	MTX (DIT)		630-2500A			
Dimension (WxDxH,mm)		1200x2505x2400	1200x2505x2400	1200x2805x2400	1200x2805x2400	1200x2805x2400
Main Apparatus	HVX/FP/MTX	1		1	1	
	CT	3(2)	3(2)	3(2)	3(2)	3(2)
	PT				3	
	ES	1	1	1	1	1
	SA	3	3	3	3	3
	High Voltage Fuse				3	
	Voltage Indicator	1	1	1	1	1
No.		I/F06	I/F07	I/F08	I/F09	
						
Rated Current	HVX (VCB)				630-3150A	
	FP (SF6)				630-2000A	
	MTX (DIT)	630-2500A	630-2500A	630-2500A		
Dimension (WxDxH,mm)		1200x2805x2400	1200x2805x2400	1200x2805x2400	1200x2805x2400	
Main Apparatus	HVX/FP/MTX	1			1	
	CT	3(2)	3(2)	3(2)	6	
	PT		3			
	ES	1	2	2	2	
	SA	3	3	3	3	
	High Voltage Fuse		3			
	Voltage Indicator	1	1	1	1	

Primary Scheme

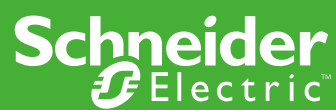
No.		P01	P02	P03	P04
Voltage Measurement/Metering					
Dimension (WxDxH,mm)		1200x2505x2400	1200x2505x2400	1200x2505x2400	1200x2505x2400
Main Apparatus	CT				
	PT	3	3	3	3
	ES			1	1
	SA	3	3	3	3
	High Voltage Fuse	3	3	3	3
	Voltage Indicator	1	1	1	1

DNF7 Metal-Clad Switchgear

Primary Scheme

No.		CR 01	CR 02	
				
Rated Current	VCB (HVX)		630-2500A	
	FP (SF6)		630-2500A	
	MTX (DIT)	630-2500A		
Dimension (WxDxH,mm)		1200x2505x2400	1200x2505x2400	
Main Apparatus	HVX/FP/MTX	1	1	
	CT	3		
	PT			
	ES			
	SA			
	High Voltage Fuse			
	Voltage Indicator	1		
No.		CR 03	CR 04	CR 05
				
Rated Current	HVX (VCB)	630-3150A	630-3150A	630-3150A
Dimension (WxDxH,mm)		1200x2505x2400		
Main Apparatus	HVX/FP/MTX			
	CT	3		
	PT	3		
	ES	1		
	SA			
	High Voltage Fuse	3		
	Voltage Indicator	1		

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ONE0006
2017.08

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